

Clinical Decision Making and Pattern Recognition

Agentic AI, Classification, Prediction, and Anomaly Detection for Treatment, Payment, and Operations

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Submitted to Cotiviti | June 2025

Defining the Concept

Clinical decision making is the process by which clinicians, administrators, and payers interpret complex, heterogeneous patient and operational data to select appropriate courses of action. Historically reliant on physician judgment and static rule sets, this domain is undergoing a fundamental transformation as machine learning (ML), artificial intelligence (AI), and large language models (LLMs) converge with electronic health record (EHR) infrastructure and real-time data streams.

Pattern recognition the computational identification of clinically meaningful signals within high-dimensional data—serves as the technical backbone of modern clinical AI. Key methodologies include supervised classification (e.g., logistic regression, gradient boosting, neural networks), predictive modeling for outcomes and resource utilization, unsupervised clustering for patient phenotyping and fraud detection, time-series anomaly detection for vital sign surveillance and billing irregularities, and chain-of-thought or agentic reasoning frameworks in which LLMs decompose multi-step clinical or administrative problems autonomously.

Together, these capabilities address Cotiviti's core domain—Treatment, Payment, and Operations (TPO)—by enabling evidence-based care escalation, automated payment integrity review, and intelligent operational workflow optimization.

Relevant Trends

Generative AI and Agentic Architectures

The emergence of LLMs such as GPT-4 and Claude has catalyzed a shift from narrow predictive models toward agentic AI systems capable of multi-step reasoning, tool use, and autonomous task completion. In healthcare, agentic frameworks are being applied to prior authorization review, clinical documentation synthesis, and denial management—tasks that previously required significant human labor. Foundation model fine-tuning on clinical corpora (e.g., Med-PaLM 2, BioMedLM) is accelerating deployment readiness in regulated environments (Singhal et al., 2023).

Real-Time Risk Stratification

Health systems and payers are increasingly deploying ML models at the point of care and at the point of claim adjudication. CMS-aligned risk models, sepsis early warning systems, and readmission predictors are now embedded in major EHR platforms. Explainable AI techniques such as SHAP (SHapley Additive exPlanations) are becoming standard requirements for clinical adoption, enabling clinicians and auditors to understand the drivers behind any given model output (Lundberg & Lee, 2017).

Anomaly Detection for Payment Integrity

Unsupervised and semi-supervised anomaly detection algorithms are increasingly central to healthcare payment integrity workflows. Time-series and graph-based methods can identify aberrant billing patterns, upcoding, unbundling, and duplicate claims at scale—capabilities that align directly with Cotiviti's core business in retrospective and prospective payment review. The global healthcare fraud detection market is projected to reach \$11.1 billion by 2032 (MarketsandMarkets, 2023).

Regulatory and Trust Landscape

The FDA's AI/ML action plan and ONC interoperability rules are shaping deployment constraints. Algorithmic transparency, bias auditing, and clinical validation requirements are raising the bar for commercial AI in healthcare. Organizations that build explainability and audit trails into their AI architectures from inception will maintain a durable competitive advantage as regulatory scrutiny intensifies.

Opportunities and Threats

Opportunities

- Payment integrity at scale: AI-powered anomaly detection can reduce false positive audit rates while increasing detection of sophisticated fraud schemes, directly protecting payer margins.
- Autonomous prior authorization: Agentic LLM pipelines can review clinical documentation, apply payer-specific criteria, and generate determination rationales—reducing administrative cost and decision latency.
- Predictive population health: Classification and clustering models applied to claims and EHR data enable proactive identification of high-risk members before costly interventions occur.
- Explainability as a differentiator: SHAP-integrated model outputs enable auditor trust, regulatory compliance, and physician engagement in ways that black-box systems cannot.

Threats

- Model drift and validation burden: Clinical AI models degrade over time as care patterns, coding practices, and population demographics shift. Continuous monitoring infrastructure is operationally expensive.
- Adversarial gaming: As AI-based fraud detection becomes more prevalent, bad actors adapt. Providers may adjust billing patterns to evade algorithmic detection, requiring continuous model retraining.
- Regulatory uncertainty: Pending FDA AI guidance and state-level algorithmic transparency laws create compliance complexity for commercial AI deployments.
- Equity and bias risk: Models trained on historically biased claims data may systematically disadvantage certain populations, creating legal and reputational exposure.

Strategic Recommendations for Cotiviti

Based on the foregoing analysis, the following strategic investments are recommended:

1. Agentic AI for Payment Integrity Automation

Cotiviti should invest in an agentic AI layer that orchestrates LLM reasoning over clinical documentation, coding guidelines, and payer contracts to automate retrospective and prospective payment review. This architecture would integrate a classification model for claim-level anomaly

scoring, a chain-of-thought LLM for determination rationale generation, and a SHAP-based explainability module for auditor review. A phased pilot targeting high-volume DRG categories with known coding variability (e.g., sepsis, cardiac procedures) would generate measurable ROI within 12 months.

2. Real-Time Clinical Risk Stratification Platform

Cotiviti should develop or acquire a clinical risk stratification capability—distinct from claims-based tools—that ingests real-time EHR data streams via FHIR APIs to predict high-cost outcomes such as readmission, ICU escalation, and treatment non-adherence. Deploying this as a SaaS offering to health system partners would expand Cotiviti's value proposition upstream from payment into care management, creating new revenue streams and deeper payer-provider integration.

3. Federated Learning for Multi-Payer Anomaly Detection

Given that fraud and abuse patterns often span multiple payers, Cotiviti is uniquely positioned to develop a federated learning consortium in which anomaly detection models are trained across payer datasets without centralizing sensitive claims data. This approach would improve model generalizability, reduce individual payer data requirements for model training, and establish Cotiviti as the infrastructure layer for cross-industry payment integrity intelligence.

References

- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems*, 30. <https://proceedings.neurips.cc/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html>
- MarketsandMarkets. (2023). Healthcare fraud detection market by offering, delivery mode, end-user and region—Global forecast to 2032. <https://www.marketsandmarkets.com/Market-Reports/healthcare-fraud-detection-market.html>
- Obermeyer, Z., & Emanuel, E. J. (2016). Predicting the future—Big data, machine learning, and clinical medicine. *New England Journal of Medicine*, 375(13), 1216–1219. <https://doi.org/10.1056/NEJMp1606181>
- Office of the National Coordinator for Health Information Technology. (2023). *ONC's cures act final rule*. U.S. Department of Health & Human Services. <https://www.healthit.gov/curesrule/>
- Singhal, K., Azizi, S., Tu, T., Mahdavi, S. S., Wei, J., Chung, H. W., Scales, N., Tanwani, A., Cole-Lewis, H., Pfohl, S., Papadimitriou, P., Kelber, J., Shepherd, J., Kelly, F., Seneviratne, M., Dean, J., Corrado, G. S., Matias, Y., & Natarajan, V. (2023). Large language models encode clinical knowledge. *Nature*, 620, 172–180. <https://doi.org/10.1038/s41586-023-06291-2>
- U.S. Food and Drug Administration. (2021). Artificial intelligence/machine learning (AI/ML)-based software as a medical device (SaMD) action plan. <https://www.fda.gov/media/145022/download>
- Wornow, M., Xu, Y., Hegde, R., Shah, N., & Callahan, A. (2023). The shaky foundations of large language models and foundation models for electronic health records. *npj Digital Medicine*, 6(1), 135. <https://doi.org/10.1038/s41746-023-00879-8>